

Test Report: Storage & Filesystem (Cluster 2)

1. Architectural Approach

- **Method:** Zabbix Agent 2 (Active Checks) with **Filtered Low-Level Discovery (LLD)**.
- **Justification:** Filesystems are highly dynamic. Hardcoding paths like `/` or `/data` guarantees gaps in monitoring if administrators add new drives. By utilizing LLD filtered strictly to `ext4`, `xfs`, and `btrfs`, the Zabbix agent dynamically discovers and monitors all physical partitions while safely ignoring noisy virtual filesystems (e.g., `tmpfs`).

2. Implemented OS-Level Requirements

The following 3 requirements were implemented directly into the OS-level YAML template utilizing the LLD approach:

1. **[OBS-F-021] Filesystem Capacity Trend Monitoring:** Monitors raw Space Utilisation (`vfs.fs.size`) and uses Zabbix's advanced `timeleft()` function to predict if the drive will fill up within the next 7 days.
2. **[OBS-F-022] Inode Utilisation Monitoring:** Monitors index record exhaustion (`vfs.fs.inode`) to prevent write failures.
3. **[OBS-F-023] Filesystem Availability Monitoring:** Monitors `/proc/mounts` globally. If any physical drive crashes and remounts as Read-Only (`ro`), a CRITICAL alert fires instantly.

3. Initial Discovery (LLD Proof)

(The Latest Data screen below validates that the LLD successfully detected the default `/` and `/boot` physical drives and dynamically cloned the rules for each).

Subfilter affects only filtered data

HOSTS

Rhel9-Not-Hardened 7

DATA

With dataWithout data

☐	Host	Name	Last check	Last value	Change	Tags	Actions	Info
☐	Rhel9-Not-Hardened	FS /: Inode Utilisation	54s	0.5899 %			Graph	
☐	Rhel9-Not-Hardened	FS /: Predicted Time to Full	4m 56s	5.7E+300y	+5.7E+300y		Graph	
☐	Rhel9-Not-Hardened	FS /: Space Utilisation	58s	12.0424 %	+0.001692 %		Graph	
☐	Rhel9-Not-Hardened	FS /boot: Inode Utilisation	53s	0.08049 %			Graph	
☐	Rhel9-Not-Hardened	FS /boot: Predicted Time to Full	4m 55s	21028531669y	-5.7E+300y		Graph	
☐	Rhel9-Not-Hardened	FS /boot: Space Utilisation	57s	26.3749 %			Graph	
☐	Rhel9-Not-Hardened	Global: Mount Options (Read-Only Check)	3s	proc /proc proc rw,nosuid,nodev,noexec,relatime 0 0 sy...			History	

0 selected

Displaying 7 of 7 found

4. Dynamic Filesystem Detection (New Disk Validation)

To validate that the LLD architecture scales dynamically, I attached a new 50GB virtual disk to the OS and partitioned it into two separate drives using different architectures:

- **Partition 1 (/data1):** A standard `xf`s filesystem.
- **Partition 2 (/data2):** A complex Logical Volume (LVM).

Result: The LLD scanner automatically detected both new drives without requiring any modifications to the Zabbix template.

Subfilter affects only filtered data

HOSTS
Rhel9-Not-Hardened 13

DATA
With data Without data

☐	Host	Name ▲	Last check	Last value	Change	Tags	Actions	Info
☐	Rhel9-Not-Hardened	FS /: Inode Utilisation	24s	0.59 %	+0.000007 %		Graph	
☐	Rhel9-Not-Hardened	FS /: Predicted Time to Full	8m 26s	2y	-5.7E+300y		Graph	
☐	Rhel9-Not-Hardened	FS /: Space Utilisation	28s	12.0419 %	-0.000206 %		Graph	
☐	Rhel9-Not-Hardened	FS /boot: Inode Utilisation	23s	0.08049 %			Graph	
☐	Rhel9-Not-Hardened	FS /boot: Predicted Time to Full	8m 25s	5.7E+300y	+5.7E+300y		Graph	
☐	Rhel9-Not-Hardened	FS /boot: Space Utilisation	27s	26.3749 %			Graph	
☐	Rhel9-Not-Hardened	FS /data1: Inode Utilisation	20s	0.000023 %			Graph	
☐	Rhel9-Not-Hardened	FS /data1: Predicted Time to Full	1m 21s	-1s			Graph	
☐	Rhel9-Not-Hardened	FS /data1: Space Utilisation	22s	0.8236 %			Graph	
☐	Rhel9-Not-Hardened	FS /data2: Inode Utilisation	17s	0.000024 %			Graph	
☐	Rhel9-Not-Hardened	FS /data2: Predicted Time to Full	18s	-1s			Graph	
☐	Rhel9-Not-Hardened	FS /data2: Space Utilisation	19s	0.829 %			Graph	
☐	Rhel9-Not-Hardened	Global: Mount Options (Read-Only Check)	33s	proc /proc: proc rw,nosuid,nodev,noexec,relatime 0 0 s...			History	

0 selected

Displaying 13 of 13 found

5. Filesystem Availability Trigger Validation (Read-Only Test)

To validate that the global mount-options check satisfies **[OBS-F-023]**, I manually forced one of the active data drives into a `Read-Only` state.

- **Action:** Executed `mount -o remount,ro /data1` on the OS.
- **Result:** Zabbix immediately detected the state change in `/proc/mounts` and fired the **CRITICAL** threshold alert.

Problems

Export to CSV

Show Recent problems Problems History

Host groups type here to search Select

Hosts Rhel9-Not-Hardened X type here to search Select

Triggers type here to search Select

Problem

Severity ☐ Not classified ☐ Warning ☐ High ☐ Information ☐ Average ☐ Disaster

Age less than 14 days

Host inventory Type Remove

Tags And/Or Or tag Contains value Remove

Show symptoms ☐

Show suppressed problems ☐

Acknowledgement status All Unacknowledged Acknowledged By me

Save as Apply Reset

☐	Time ▼	Severity	Recovery time	Status	Info	Host	Problem	Duration
☐	07:43:00...	High				Rhel9-Not-Hardened	[OBS-F-023] CRITICAL: A physical filesystem is mounted Read-Only	1m 40s

0 selected

Displaying 1 of 1 found

Mass update

6. Architecturally Justified Requirements

Because this Proof of Concept is hosted on a Virtual Machine, it evaluates OS-level software rather than physical hardware. The remaining requirements will be monitored at the Bare-Metal or API layer in production:

- **[OBS-F-028] Logical Volume Monitoring (LVM):** Zabbix does not possess a native Go plugin for LVM metadata. This must be satisfied via a secure `UserParameter` invoking the `lvs` command.
- **[OBS-F-025] & [OBS-F-026] Storage Media Health & Redundancy:** VMs abstract SMART controllers and RAID arrays. This must be monitored directly at the Hypervisor (Proxmox/SAN) layer.
- **[OBS-F-027] Storage Latency:** VM kernels (VirtIO) often abstract physical disk read/write millisecond times. Accurate storage latency must be polled from the Bare-Metal Hypervisor.
- **[OBS-F-029], [OBS-F-030], & [OBS-F-031] Backup Operations:** An OS agent cannot know if an external enterprise Backup Server (e.g., Veeam) successfully backed it up. This is solved via Zabbix Trapper (push APIs from the Backup system).